Mitigating Emotional Inconsistency in Audio-Visual Depression Detection via Uncertainty-aware Fusion

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Abstract. Depression detection focuses on individuals' long-term and stable states, while emotional expression is typically a short-term, context-sensitive transient response. Direct fusion of emotional features with depressive features tends to misclassify "momentary emotional fluctuations" as "long-term depressive signals", thereby leading to false detection and poor generalization. To address this issue, this paper proposes a lightweight uncertainty-aware fusion framework, namely A2F-Net. Our core idea is straightforward: first, a depression branch generates a reference judgment indicating whether the current sample is more likely depressive or non-depressive. Subsequently, emotional cues are examined moment-by-moment to verify their consistency with this reference. For segments where emotional evidence is significantly inconsistent with the reference, their weights are reduced to retain only emotional information that is more conducive to depression assessment. Finally, one-way residual learning is adopted to supplement the filtered emotional information into the depression representation, preventing short-term emotional noise from interfering with the long-term depression representation. Experiments on LMVD, D-vlog, and DAIC-WOZ demonstrate that A2F-Net consistently improves depression detection performance, delivering consistent improvements over representative methods (up to 1.39% in F1-score) and exhibiting stronger robustness under intense emotional fluctuations.

Keywords: Depression Detection, Emotional Inconsistency, Uncertainty-aware Fusion, Multimodal Learning.

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1 Introduction

Depression is one of the most severe public health challenges of the 21st century, affecting more than 280 million people worldwide. The World Health Organization (WHO) predicts that by 2030, depression will become the leading contributor to the global disease burden. Conventional clinical diagnosis primarily relies on subjective instruments such as self-report questionnaires and structured interviews. Although clinically meaningful, these approaches have practical limitations in terms of objectivity, scalability, and early screening efficiency: the outcomes can be influenced by self-report bias and contextual factors, and may be distorted when patients conceal symptoms. To address these issues, recent research has increasingly focused on automatic depression detection based on objective behavioral cues. Audio-visual modalities can capture depression-relevant

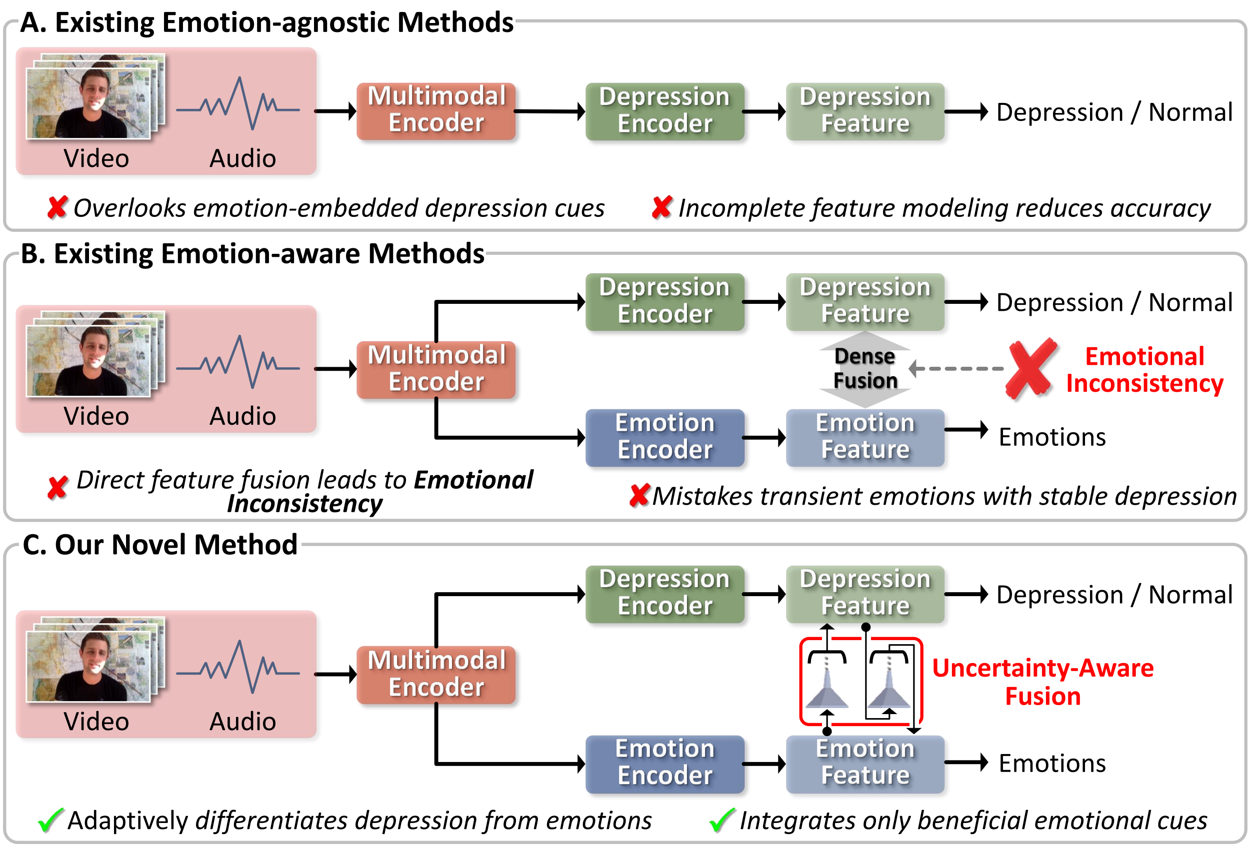

Fig 1: Comparison of existing and proposed methods for audio-visual depression detection. (A) Emotion-agnostic methods overlook emotion-embedded depressive cues, leading to incomplete modeling. (B) Direct fusion of emotional and depressive features can confuse topic-driven transient moods with stable depressive evidence, leading to representation conflicts and misclassification. (C) Our proposed A2F-Net performs uncertainty-aware fusion by assessing the reliability of emotional cues before integration, suppressing high-conflict segments, and injecting reliable cues into the depressive representation in a controlled manner.

behavioral manifestations from facial expressions,[1] speech prosody,[2] and body movements,[3] and have therefore attracted substantial attention in multimodal depression detection. With the advance of deep learning, the research paradigm has gradually shifted from unimodal analysis to multimodal fusion, aiming to integrate complementary information sources to model complex affective and behavioral patterns.

Existing audio-visual depression detection methods[4–7] can be broadly categorized into two types (see Fig. 1). The first type, emotion-agnostic methods (Fig. 1-A), tends to downplay or avoid explicit emotional cues and instead focuses on more stable depression-related biomarkers, such as slowed expressiveness, abnormal speech rhythm, gaze avoidance, and flattened affect. While this strategy avoids an explicit reliance on transient emotional cues and encourages modeling more stable depression-related biomarkers, it also has an evident limitation: depression is not merely a reduction in emotion, but can systematically alter emotional reactivity and expressive strategies, such as blunted responses to positive stimuli and context-incongruent micro-expressions. Completely excluding emotion-related information may therefore miss behavior changes modulated by depression, resulting in incomplete representations. The second type, emotion-aware methods (Fig. 1-B), is motivated by the assumption that "depression alters typical affective patterns", and explicitly incorporates emotional features for joint modeling with depressive cues. Although this line of work can offer stronger representational capacity, it is also more vulnerable to short-term emotional fluctuations, particularly when interview topics and interaction styles vary; overly direct fusion strategies may consequently introduce greater uncertainty.[8–10] In this paper, *uncertainty* refers to the *inconsistency risk* of segment-level emotional evidence with respect to the sample-level depressive state, i.e., how likely a local cue is driven by situational factors and thus unreliable for long-term assessment.

A key challenge lies in the temporal-scale mismatch between emotion and depression. Emotion is short-term and highly context-dependent, and the same subject may fluctuate markedly with topics, interaction patterns, or even momentary memory recall. In contrast, depression detection targets a long-term and relatively stable state. In real interviews, it is common that a non-depressed subject temporarily feels low due to a certain topic, and a depressed subject may appear positive or engaged in some segments. If a model treats such segment-level emotions as stable evidence and fuses them indiscriminately, it can introduce conflicts at the representation level, leading to false

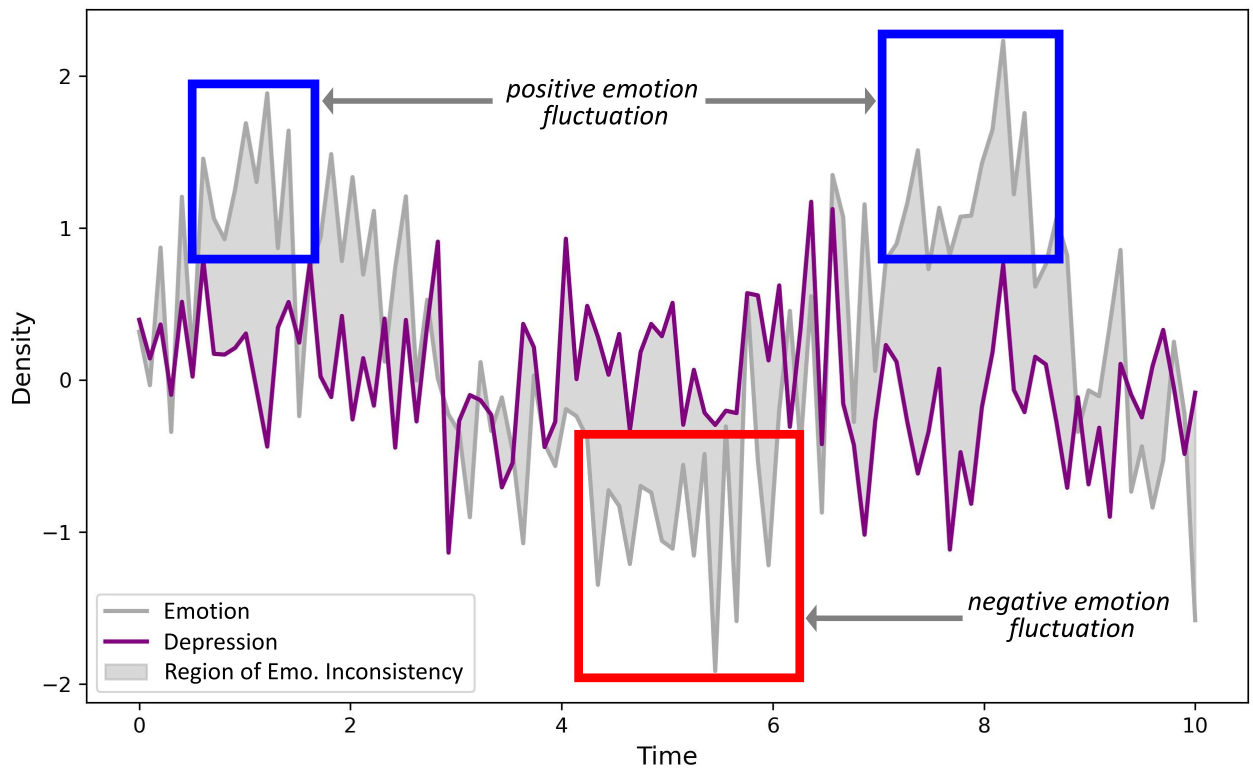

Fig 2: Visualization of emotional inconsistency. This figure illustrates the overlap between transient emotional fluctuations (positive and negative) and stable depressive signals over time. The shaded region represents the area of *emotional inconsistency*, where emotional fluctuations complicate the accurate identification of depressive symptoms.

positives and false negatives. We refer to this phenomenon as *emotional inconsistency*, i.e., "local emotional evidence that is inconsistent with the global depressive state and contaminates stable depressive cues during fusion" (see the conceptual illustration in Fig. 2). Therefore, a central challenge is to distinguish emotional expressions that reflect depressive pathological traits from those driven by situational factors, namely, to decide before fusion whether "the emotion in this segment provides depression-relevant evidence or introduces depression-irrelevant noise".

To address the representation conflicts caused by emotional inconsistency, we propose A2F-Net (Fig. 1-C), a lightweight uncertainty-aware fusion framework. It explicitly learns to down-weight unreliable emotional evidence while preserving stable depressive cues. Accordingly, *uncertainty-aware* means explicitly estimating this risk and using it to down-weight high-risk segments before injecting emotional information into the depressive representation. Although uncertainty-aware fusion is a common idea in multimodal learning, the uncertainty considered here is task-specific. In depression detection, emotional expressions are short-term and topic-dependent, whereas depression is long-term and relatively stable. Therefore, a clear emotional cue may still be unreliable if it conflicts with the sample-level depressive tendency. This motivates us to estimate uncertainty as the consistency risk between local emotional evidence and the global depressive state.

Specifically, A2F-Net introduces three complementary components to achieve this goal. First, an uncertainty-aware temporal gating mechanism selectively suppresses emotionally volatile segments that are likely to contradict the sample-level depressive state, thereby reducing the risk of treating transient affect as persistent depressive evidence. Second, a one-way residual fusion scheme integrates the screened emotional information as a controlled complement to the depressive representation, preventing short-term affective noise from overriding the long-term carrier. Third, a long-term stability regularization is incorporated to encourage temporally stable depressive representations and discourage redundancy between the main depressive carrier and the injected emotional increment. We conduct extensive experiments on public datasets including LMVD,[11] D-vlog,[12] DAIC-WOZ,[13] and the results show that A2F-Net achieves consistent performance gains with low computational overhead, while exhibiting stronger robustness and clearer interpretability on samples with pronounced emotional fluctuations.

In summary, the main contributions of this paper are as follows:

- We identify and formalize emotional inconsistency arising from the temporal-scale mismatch between emotion and depression, and analyze how it induces representation conflicts in audio-visual fusion for depression detection.

- We propose a lightweight uncertainty-aware fusion framework, A2F-Net, which explicitly screens time-local emotional evidence by estimating its inconsistency risk with the sample-level depressive state.
- We design two effective components to realize robust fusion, including an uncertainty-aware temporal gating strategy to down-weight high-risk segments and a one-way residual fusion scheme to inject screened emotional cues as a controlled complement while preserving stable depressive representations.
- We introduce a long-term stability regularization to encode the slow-varying nature of depression and validate consistent improvements in accuracy, robustness, and interpretability on LMVD, D-vlog, and DAIC-WOZ.

2 Related Works

Automatic depression detection (ADD) has progressed rapidly with audio-visual and increasingly linguistic signals. Benchmarks such as AVEC'13/14 and AVEC'17 established protocols for affect and depression estimation and catalyzed multimodal pipelines.[13–15] Beyond controlled interviews, in-the-wild resources such as D-vlog[12] and large-scale datasets like LMVD[11] improve robustness under real-world conditions, while semi-structured clinical corpora support screening-oriented research.[16] Standard acoustic toolkits (openSMILE[17]) and early deep audio models (DepAudioNet[18]) provide strong baselines, and given the prevalence of clinical scales, label reliability concerns such as BDI motivate careful protocol design.[19]

2.1 *Emotion-agnostic Audio-visual Methods*

Many studies model paralinguistic and nonverbal markers of depression without explicit emotion supervision. On the acoustic side, CNN and LSTM encoders on spectrograms or raw waveforms, along with spatiotemporal stacks, capture prosodic and articulation cues of depressive speech.[18,20] Self-supervised speech encoders such as wav2vec 2.0 and HuBERT further improve robustness under limited labels.[21] On the visual side, deep networks model facial appearance, optical flow, 3D dynamics, and multi-scale temporal differences to detect diminished expressivity and altered micro-dynamics.[22–24] Body-level signals, including self-adaptors and fidgeting, provide complementary cues.[25] Multimodal variants fuse audio-visual streams using hierarchical or bridge-style strategies while remaining emotion-agnostic.[26,27] These pipelines are effective but may miss higher-level affective context and nuanced depressive patterns.

2.2 *Emotion-aware Audio-visual Methods*

To capture depression-affect interactions, emotion-aware methods incorporate emotion supervision using auxiliary labels, intermediate affect representations, or attention to emotionally salient segments. Temporal transformers jointly learn depression and emotion,[28] and multi-task frameworks integrate sentiment or affect cues in multimodal fusion.[29] Some approaches explicitly track emotional changes across modalities to stabilize predictions.[30] However, uniformly leveraging emotional cues may conflate transient affect such as brief joy or sadness with stable depressive traits like anhedonia, producing *Emotional Inconsistency*. Evidence arises from uncertainty modeling for affect annotation[31] and modality-specific observations in depressed facial behavior.[32] In speech-text settings, cross-modal attention can bias depression features if contextual or stylistic information is not disentangled.[33] Consequently, naive emotion integration may add noise and weaken robustness, especially for long weakly labeled sequences in interviews and vlogs.

2.3 *Uncertainty-aware Fusion*

In long-form interviews and vlogs, affective signals are often noisy and context-dependent, and segment-level emotional expressions may contradict the slowly-varying depressive state. This has motivated uncertainty-aware modeling in affective computing, where ambiguity and annotation uncertainty are explicitly considered to improve robustness.[34–36] In depression detection, related attempts typically rely on attention or auxiliary affect objectives to handle noisy emotional cues, and some methods track affective changes to stabilize predictions.[37,38] However, most existing designs[39–41] do not explicitly quantify whether a segment-level emotional cue is *consistent* with the sample-level depressive state, and symmetric fusion may still overwrite long-term depressive representations with short-term fluctuations under topic shifts and emotional volatility.

Most uncertainty-aware fusion methods focus on modality reliability, noisy inputs, or ambiguous annotations. In contrast, A2F-Net focuses on whether segment-level emotional evidence should be trusted for depression assessment. We therefore define uncertainty as anchor-consistency risk and use it to suppress emotional cues that may reflect transient context rather than stable depression.

Our work addresses this gap by treating uncertainty as the anchor-consistency risk of local emotional evidence and designing uncertainty-aware temporal gating with one-way residual injection to suppress high-conflict segments while preserving the long-term carrier.

3 The Proposed Method

3.1 Overall Framework

A2F-Net targets binary depression detection from an aligned audio-visual interview sequence $\mathcal{X} = \{(x_t^v, x_t^a)\}_{t=1}^{T}$. As shown in Fig. 3, we first perform shared modality encoding to obtain a segment-level multimodal representation sequence F, and the depression branch produces a sample-level prediction $\hat{y}_d$ as a global state anchor, indicating whether the current sample is more likely depressive or non-depressive. Instead of directly fusing emotional cues into the depressive carrier, A2F-Net estimates segment-wise emotional evidence and applies an uncertainty-aware temporal gating (Sec. 3.3) to evaluate its consistency with the anchor, actively suppressing emotionally volatile segments that contradict the global state and thus reducing the risk of mistaking transient moods as long-term depressive signals. The screened emotional information is then injected into the depression representation via one-way residual fusion (ORF) (Sec. 3.4), so that emotional cues act as a controlled increment rather than overwriting the long-term carrier. During training, a long-term stability regularization (LTSR) (Sec. 3.5) further encourages slowly-varying depressive representations and discourages redundancy between the main carrier and the injected increment. Finally, temporal aggregation yields the prediction $\hat{y}$, optimized primarily with binary cross-entropy.

3.2 Shared Encoding and Dual-Branch Temporal Representations

Given an interview recording of duration S seconds, we partition it into T temporally aligned segments with a fixed segment length L (seconds), where

$$T = \left\lceil \frac{S}{L} \right\rceil . \tag{1}$$

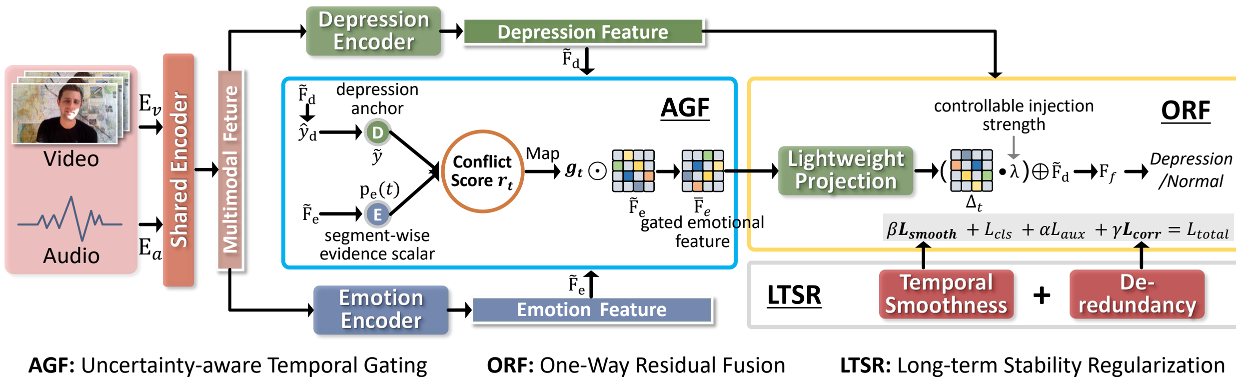

Fig 3: Overall framework of A2F-Net: shared audio-visual encoding produces multimodal features; anchor-guided uncertainty-aware temporal gating (AGF, Sec. 3.3) suppresses emotion segments conflicting with the depression anchor; one-way residual fusion (ORF, Sec. 3.4) injects reliable emotional increments into the depression representation; long-term stability regularization (LTSR, Sec. 3.5) with temporal smoothness and de-redundancy constraints further improves robustness for binary depression detection

For the t-th segment, we uniformly sample K video frames and extract the corresponding audio clip of length L:

$$x_t^v = \{I_{t,1}, \dots, I_{t,K}\}, \qquad x_t^a \in \mathbb{R}^{L \cdot f_s}, \tag{2}$$

where f_s denotes the audio sampling rate. A video encoder $E_v(\cdot)$ (e.g., Swin-B[42]) produces frame-level embeddings which are aggregated into a segment-level visual representation, and an audio encoder $E_a(\cdot)$ (e.g., WavLM[43]) produces a segment-level acoustic representation:

$$z_{t,k}^v = E_v(I_{t,k}) \in \mathbb{R}^{D_v}, \quad h_t^v = \mathrm{Pool}_v(\{z_{t,k}^v\}_{k=1}^K) \in \mathbb{R}^{D_v}, \quad h_t^a = E_a(x_t^a) \in \mathbb{R}^{D_a}. \tag{3}$$

We then lightly fuse the two modalities to obtain a unified multimodal segment embedding

$$f_t = \phi([h_t^v; h_t^a]) \in \mathbb{R}^D, \tag{4}$$

forming $F = \{f_t\}_{t=1}^{T} \in \mathbb{R}^{B\times T\times D}$, where B is the batch size and $\phi(\cdot)$ is a lightweight two-layer MLP (Linear–ReLU–Linear).

Since interview lengths vary across subjects, we pad the segment sequence to a fixed maximum length $T_{\max}$ within each mini-batch and use a binary mask $M \in \{0,1\}^{B\times T_{\max}}$ to mark valid segments (1 for valid, 0 for padded), so that temporal models and pooling ignore padded positions:

$$F \in \mathbb{R}^{B\times T_{\max}\times D}, \qquad M \in \{0,1\}^{B\times T_{\max}}. \tag{5}$$

Two branches share F but operate in different projected subspaces with independent temporal modeling: the depression branch captures slowly-varying long-term patterns to output a sample-level anchor, while the emotion branch captures rapidly-varying local cues to output segment-wise evidence for consistency evaluation:

$$F_d = \psi_d(F),\;\; \hat{y}_d = \sigma\big(\mathrm{Pool}(T_d(F_d; M))\big),\;\; F_e = \psi_e(F),\;\; p_e(t) = \sigma\big(w_e^{\top} T_e(F_e; M)_t + b_e\big). \tag{6}$$

Here $T_d(\cdot)$ can be implemented by a lightweight Transformer/GRU/TCN with $\mathrm{Pool}(\cdot)$ (e.g., attention pooling) to obtain a sequence-level representation, while $T_e(\cdot)$ is instantiated with a lighter GRU or 1D-Conv to produce segment-wise context. The mask M is applied to prevent padded segments from contributing to temporal modeling and aggregation.

We treat $p_e(t) \in (0,1)$ as a comparable scalar indicating the direction of emotional evidence for subsequent anchor-consistency gating rather than an explicit emotion category prediction. The label is $y \in \{0,1\}$, where $y = 1$ indicates depression.

3.3 Uncertainty-aware Temporal Gating

Motivation. Depression detection aims at estimating a long-term and stable state at the sample level, whereas emotional expressions are short-term, topic-dependent, and often highly volatile in interviews. A non-depressed subject may become temporarily low on certain questions, and a depressed subject may appear engaged in some segments; thus local emotional evidence is not always consistent with the global depressive state. If emotional cues are injected into the depressive carrier without discrimination, transient fluctuations can be mistaken as persistent depressive signals, leading to representation contamination and degraded generalization. To address this issue, we propose **Uncertainty-aware Temporal Gating (AGF)**: it uses the depression-branch anchor prediction $\hat{y}_d$ as a global reference, evaluates the consistency of segment-wise emotional evidence $p_e(t)$ against this anchor, and adaptively down-weights high-conflict segments before fusion (Fig. 3).

Technical Details. AGF uses a global anchor to constrain local evidence and produces an interpretable temporal weight g_t. We first construct a binary anchor label from the depression-branch prediction $\hat{y}_d$ and stop its gradient to prevent the gating module from back-propagating adversarial signals to the anchor and destabilizing training:

$$\tilde{y} = \mathbb{I}[\hat{y}_d > 0.5], \quad \tilde{y} \in \{0, 1\}, \quad \mathrm{stopgrad}(\tilde{y}). \tag{7}$$

The emotion branch outputs a segment-wise evidence scalar $p_e(t) \in (0, 1)$, indicating whether the local emotional cue tends to support the depressive (1) or non-depressive (0) direction. To quantify the inconsistency risk between local evidence and the global anchor, we define a segment-level conflict probability r_t. When $\tilde{y} = 1$, a segment that supports non-depression should yield a larger conflict, and vice versa for $\tilde{y} = 0$. This can be unified as:

$$r_t = p_e(t) \cdot (1 - \tilde{y}) + \big(1 - p_e(t)\big) \cdot \tilde{y}, \quad r_t \in (0, 1). \tag{8}$$

A larger r_t indicates higher uncertainty (i.e., stronger inconsistency) of the emotional cue with respect to the anchored global state. We then map r_t to a gating weight $g_t \in (0, 1)$ using a sigmoid with a margin τ and temperature s to control tolerance and smoothness:

$$g_t = \sigma\left(\frac{0.5 + \tau - r_t}{s}\right). \tag{9}$$

Here, the margin τ shifts the suppression threshold to $0.5 + \tau$. A larger τ provides higher tolerance for emotional inconsistency, penalizing the segment (i.e., $g_t < 0.5$) only when its conflict risk r_t exceeds this boundary. The temperature s controls the steepness of the gating transition, balancing a hard binary selection with smooth gradient flow during training. In our experiments, we empirically set $\tau = 0.1$ and $s = 0.05$ based on validation performance.

Finally, we apply g_t to the emotion temporal representation. Let $\tilde{F}_e = T_e(F_e) \in \mathbb{R}^{B \times T \times D_e}$ denote the contextualized emotion features, then the gated emotional features are:

$$\bar{F}_{e,t} = g_t \odot \tilde{F}_{e,t}. \tag{10}$$

The resulting $\bar{F}_e$ is fed to the subsequent fusion module, and g_t directly serves as an interpretable segment-wise reliability weight.

Summary. AGF explicitly quantifies emotional uncertainty via the anchor-consistency conflict r_t and generates interpretable temporal gates g_t to suppress high-risk segments, reducing representation contamination before fusion. Despite being lightweight, it provides cleaner and more stable emotional inputs for the subsequent one-way residual fusion. The depression anchor provides a pre-fusion estimate of the long-term depressive tendency. AGF uses this anchor to suppress emotional segments that conflict with the global state. ORF injects the screened emotional information as a bounded residual complement, preventing emotion from overwriting the depression carrier. LTSR further encourages the carrier to follow the slow-varying nature of depression.

3.4 One-Way Residual Fusion

Motivation. AGF suppresses segments whose emotional evidence is highly inconsistent with the global anchor, yet a fundamental issue remains: emotional cues are inherently fast-varying local signals, often exhibiting much larger temporal fluctuations than the slowly-varying patterns relevant to depression. With symmetric fusion (e.g., concatenation followed by joint updates or bidirectional interactions between branches), the representation may still over-react to short-term emotional swings and thus undermine the stability of the depressive carrier. This is particularly problematic in interviews where topic-driven emotional responses vary drastically across segments and do not necessarily imply a change in the underlying depressive state. Therefore, we explicitly separate the *carrier* and the *complement*: the depression branch should dominate as a long-term carrier, while emotional information should only provide controlled, reliability-aware supplementation. To this end, we propose **One-Way Residual Fusion (ORF)**, which injects screened emotional cues into the depressive representation as an additive residual increment, preventing transient emotional noise from overwriting stable depressive signals (Fig. 3).

Technical Details. ORF follows a simple principle: *add without rewriting*. We treat the depression-branch temporal representation as a stable base, $\tilde{F}_d = T_d(F_d) \in \mathbb{R}^{B \times T \times D_d}$. The emotion branch yields contextualized segment features $\tilde{F}_e = T_e(F_e)$, and AGF provides temporal gates g_t to obtain gated emotion features $\bar{F}_e \in \mathbb{R}^{B \times T \times D_e}$. Since the two branches typically have different channel dimensions and emotional cues should act as a complement, we employ a lightweight projection $\rho(\cdot)$ to map gated emotional features into the depressive subspace and form a residual increment:

$$\Delta_t = \rho(\bar{F}_{e,t}) \in \mathbb{R}^{D_d}, \tag{11}$$

where $\rho(\cdot)$ is a lightweight projection layer that maps the gated emotion feature $\bar{F}_{e,t}$ into the depression feature space. The fused representation is then obtained through one-way residual injection:

The projection can be instantiated with one or two linear layers, serving two purposes: (i) aligning dimensions, and (ii) learning which emotional cues are worth being expressed as a residual patch. We then inject the increment in a residual manner to form the fused sequence F_f. To prevent the residual increment from dominating the base depressive carrier, we introduce a bounded learnable injection strength λ (see Table 14), that explicitly controls the magnitude of emotional supplementation. Instead of using an unconstrained scalar, we parameterize λ with an upper bound to ensure stable and conservative fusion. The fused sequence is then obtained by one-way residual injection:

$$F_{f,t} = \tilde{F}_{d,t} + \lambda\, \Delta_t, \quad \lambda = \lambda_{\max} \cdot \sigma(\alpha), \quad \alpha \in \mathbb{R}, \quad F_f \in \mathbb{R}^{B \times T \times D_d}. \tag{12}$$

where, $\widetilde{F}_{d,t} \in \mathbb{R}^{D_d}$ denotes the pre-fusion depression carrier at segment t, Δ_t denotes the projected emotional residual, and $F_{f,t}$ denotes the final fused representation. The scalar λ controls how much screened emotional information is injected into the depression carrier. To avoid excessive emotional interference, λ is bounded by $\lambda_{\max}$, while α is an unconstrained learnable parameter. Therefore, emotional information can only enter the depression representation as a magnitude-controlled residual complement, rather than rewriting the depression carrier.

Importantly, the "one-way" property is enforced as an information-flow constraint: emotional cues can only influence the depressive representation through the additive residual path scaled by $\lambda_{\max}$ whereas the depressive carrier $\tilde{F}_d$ is never re-encoded or rewritten by emotional signals. This design preserves the long-term stability of the carrier and bounds the influence of residual emotional noise that may survive AGF, making it less likely to distort the long-term structure encoded in $\tilde{F}_d$. Finally, F_f is fed into the same temporal aggregation module as in the overall framework to produce the final prediction $\hat{y}$.

Summary. ORF fixes the depressive representation as the long-term carrier and treats AGF-screened emotional cues as an optional residual patch, injected in a one-way and magnitude-controlled manner. This design preserves the benefit of emotion-aware supplementation while structurally suppressing the interference of short-term emotional fluctuations, leading to more stable and robust fusion.

3.5 Long-term Stability Regularization

Motivation. As a long-term state, depression should be represented by temporally smooth and slowly-varying patterns, while emotional cues are allowed to fluctuate sharply across segments. With classification supervision alone, a model may exploit high-frequency segment differences to fit labels, becoming overly sensitive to transient emotional swings and weakening long-term depression modeling. Moreover, ORF injects screened emotional information as an increment Δ_t into the depressive carrier $\tilde{F}_d$; without constraints, the increment may become highly correlated with the carrier, resulting in redundant encoding and amplifying residual emotional noise. To address these issues, we propose **Long-term Stability Regularization (LTSR)**, which explicitly encodes the long-term stability prior and discourages statistical redundancy between the injected increment and the depressive carrier with minimal overhead (Fig. 3).

Technical Details. LTSR consists of two complementary regularizers, targeting (i) temporal stability of the depressive carrier and (ii) complementarity of the injected increment.

Temporal smoothness. We impose a smoothness penalty on the depression-branch contextual sequence $\tilde{F}_d \in \mathbb{R}^{B\times T\times D_d}$ by discouraging abrupt changes between adjacent segments:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{B(T-1)} \sum_{b=1}^{B} \sum_{t=2}^{T} \left\| \tilde{F}_{d,t}^{(b)} - \tilde{F}_{d,t-1}^{(b)} \right\|_2^2. \tag{13}$$

Here, B is the batch size, T is the number of valid temporal segments, and $\widetilde{F}_{d,t}^{(b)}$ denotes the depression-branch representation of the t-th segment in the b-th sample. This term penalizes abrupt changes between adjacent depression representations. It is applied only to the depression carrier, because depression is assumed to be relatively stable over time, while the emotion branch is allowed to capture faster local fluctuations. This term suppresses unnecessary high-frequency oscillations, encouraging $\tilde{F}_d$ to capture slowly-varying and generalizable depressive patterns.

De-redundancy via cross-correlation. To discourage the ORF increment $\Delta_t = \rho(\bar{F}_{e,t})$ from copying the depressive carrier, we penalize the cross-correlation between $\tilde{F}_d$ and Δ across feature dimensions. Specifically, we reshape them into $N \times D_d$ matrices with $N = B \cdot T$: $\mathbf{Z}_d =$

Algorithm 1 Training of A2F-Net with AGF, ORF, and LTSR

Require: Aligned segments $\mathcal{X} = \{(x_t^v, x_t^a)\}_{t=1}^T$, label y
Require: Modules $E_v, E_a, \phi, \psi_d, \psi_e, T_d, T_e, \rho, \mathrm{Pool}$
Require: Hyperparams $\tau, s, \lambda, \alpha, \beta, \gamma$
for $t = 1$ to T **do**
  $h_t^v \leftarrow E_v(x_t^v); \quad h_t^a \leftarrow E_a(x_t^a)$
  $f_t \leftarrow \phi([h_t^v; h_t^a])$
end for
$F \leftarrow \{f_t\}_{t=1}^T; \quad F_d \leftarrow \psi_d(F); \quad F_e \leftarrow \psi_e(F)$
$\tilde{F}_d \leftarrow T_d(F_d)$
$\hat{y}_d \leftarrow \sigma(\mathrm{Pool}(\tilde{F}_d))$ ▷ depression anchor
$\tilde{F}_e \leftarrow T_e(F_e)$
for $t = 1$ to T **do**
  $p_e(t) \leftarrow \sigma(w_e^\top \tilde{F}_{e,t} + b_e)$
end for
$\tilde{y} \leftarrow \mathbb{I}[\hat{y}_d > 0.5]; \quad \tilde{y} \leftarrow \mathrm{stopgrad}(\tilde{y})$
for $t = 1$ to T **do** ▷ AGF + ORF
  $r_t \leftarrow p_e(t)(1 - \tilde{y}) + (1 - p_e(t))\tilde{y}$
  $g_t \leftarrow \sigma\left(\frac{0.5+\tau-r_t}{s}\right)$
  $\bar{F}_{e,t} \leftarrow g_t \odot \tilde{F}_{e,t}$
  $\Delta_t \leftarrow \rho(\bar{F}_{e,t})$
  $F_{f,t} \leftarrow \tilde{F}_{d,t} + \lambda\,\Delta_t$
end for
$\hat{y} \leftarrow \sigma(\mathrm{Pool}(F_f))$
$\mathcal{L}_{\text{cls}} \leftarrow \mathrm{BCE}(\hat{y}, y)$
$\mathcal{L}_{\text{aux}} \leftarrow \mathrm{BCE}(\hat{y}_d, y)$
Compute $\mathcal{L}_{\text{smooth}}$ and $\mathcal{L}_{\text{corr}}$ as defined in Sec. LTSR
$\mathcal{L} \leftarrow \mathcal{L}_{\text{cls}} + \alpha\mathcal{L}_{\text{aux}} + \beta\mathcal{L}_{\text{smooth}} + \gamma\mathcal{L}_{\text{corr}}$
Update parameters by back-propagation

$\mathrm{reshape}(\tilde{F}_d) \in \mathbb{R}^{N \times D_d}$ and $\mathbf{Z}_\Delta = \mathrm{reshape}(\Delta) \in \mathbb{R}^{N \times D_d}$, and apply per-dimension normalization to obtain $\hat{\mathbf{Z}}_d, \hat{\mathbf{Z}}_\Delta$ (zero mean, unit variance). We then compute the cross-correlation matrix:

$$\mathbf{C} = \frac{1}{N}\hat{\mathbf{Z}}_d^\top \hat{\mathbf{Z}}_\Delta \in \mathbb{R}^{D_d \times D_d}, \tag{14}$$

and minimize

$$\mathcal{L}_{\text{corr}} = \|\mathbf{C}\|_F^2. \tag{15}$$

Optimization Objective. We optimize A2F-Net with binary cross-entropy on the final prediction:

$$\mathcal{L}_{\text{cls}} = \mathrm{BCE}(\hat{y}, y), \quad y \in \{0, 1\}. \tag{16}$$

To make the depression-branch anchor more reliable and provide a stable reference for AGF, we apply an auxiliary supervision to the anchor prediction $\hat{y}_d$:

$$\mathcal{L}_{\text{aux}} = \mathrm{BCE}(\hat{y}_d, y). \tag{17}$$

The overall objective is:

$$\mathcal{L}_{total} = \mathcal{L}_{\text{cls}} + \alpha\,\mathcal{L}_{\text{aux}} + \beta\,\mathcal{L}_{\text{smooth}} + \gamma\,\mathcal{L}_{\text{corr}}, \tag{18}$$

where α, β, γ balance the auxiliary anchor supervision and the stability regularizers. In our implementation, we empirically set $\alpha = 0.5$, $\beta = 0.1$, and $\gamma = 0.1$ based on the validation performance. Note that AGF uses $\tilde{y} = \mathbb{I}[\hat{y}_d > 0.5]$ with $\mathrm{stopgrad}(\tilde{y})$ to avoid unstable gradients induced by thresholding. The overall training and optimization procedure is summarized in Algorithm 1.

Summary. LTSR enforces long-term stability and increment complementarity with minimal cost: the smoothness term suppresses high-frequency fluctuations of the depressive carrier, while the cross-correlation term reduces redundancy between the injected increment and the carrier. Together, they improve robustness and generalization under pronounced emotional fluctuations and frequent topic shifts.

4 Experiments

4.1 Experimental Setup, Datasets, and Metrics

Experimental Setup. A2F-Net was implemented in PyTorch and trained on a single NVIDIA RTX 4090 GPU. We used Adam as the optimizer with a learning rate of 1×10^{-5} and a batch size of 16, and trained the model for up to 300 epochs with early stopping based on the validation performance. For each interview/video, we construct an aligned audio-visual segment sequence $\mathcal{X} = \{(x_t^v, x_t^a)\}_{t=1}^{T}$ by uniformly sampling temporally aligned segments, and the model is trained end-to-end with the proposed AGF, ORF, and LTSR modules. Unless otherwise specified, all methods are evaluated under the same data split protocol as provided by the datasets and commonly used in prior studies, ensuring fair comparison. A2F-Net has 9.1M parameters and requires 1.3 GFLOPs per segment.

Datasets. We conduct experiments on three public audio-visual depression detection datasets: LMVD,[11] D-vlog,[12] and DAIC-WOZ.[13] LMVD contains 1,823 videos (214 hours) from 1,475 participants, including 908 depression and 915 non-depression samples, and provides pre-extracted behavioral cues such as facial action units, landmarks, and eye movements. D-vlog includes 961 YouTube vlogs (160 hours) from 816 speakers, with 555 depression and 406 non-depression samples. DAIC-WOZ consists of 189 interview sessions (733 minutes, about 16 minutes per session on average) collected with the virtual interviewer "Ellie" and annotated by PHQ-8 scores, providing multimodal features such as COVAREP audio descriptors and facial action units. Following common practice, we use the official labels/splits of each dataset (or the standard protocol adopted in prior work) for training and evaluation.

Metrics. Following previous studies,[44] we report four widely-used metrics for binary depression detection: Accuracy, Precision, Recall, and F1-score, where *depression* is treated as the positive class.

Statistical testing protocol. To reduce the influence of random initialization and to avoid over-interpreting small absolute differences, A2F-Net and the strongest baseline READ were repeated with five random seeds under the same official data splits. We report the mean and standard deviation of F1-score. For the comparison with READ, we further conducted a paired bootstrap significance test on the test-set predictions with 10,000 resamples, using F1-score as the primary statistic. A difference was considered statistically significant when $p < 0.05$.

4.2 Quantitative Evaluations

Table 1 reports quantitative comparisons on three public depression detection datasets, i.e., LMVD, D-vlog, and DAIC-WOZ. Overall, A2F-Net achieves the best or competitive mean performance across the three datasets. Compared with READ, the absolute gains on LMVD and DAIC-WOZ are modest, while the improvement on D-vlog is more evident. Therefore, we interpret the results not as large-margin superiority over READ, but as consistent evidence that uncertainty-aware screening and one-way residual injection can provide stable benefits. This conclusion is further supported by the component analysis and robustness studies under controlled emotional inconsistency, high emotion volatility, modality missingness, and temporal misalignment.

Results on LMVD. On LMVD, we compared with state-of-the-art methods, including SERes,[45] ViT,[46] MMD,[11] DepM[44] and READ.[47] As shown in Table 1a, A2F-Net obtains the best performance on all four metrics, reaching an accuracy of 77.93%, precision of 78.51%, recall of 77.84%, and F1-score of 77.73%. Compared with the strongest competitor READ, A2F-Net yields consistent gains of +0.34% in accuracy, +0.41% in precision, +0.25% in recall, and +0.26% in F1-score.

The larger margins over earlier representative methods (e.g., MMD) further suggest that selectively integrating reliable emotional cues can provide persistent benefits for depression assessment on LMVD.

Results on D-vlog. On D-vlog, we compared with state-of-the-art methods, including JAMFN,[48] DepMS,[49] STFT,[50] ABDM,[51] DepM[44] and READ.[47] We report precision, recall, and F1-score. As shown in Table 1b, A2F-Net achieves the best precision (74.16%), recall (79.68%), and F1-score (77.91%). In particular, compared with READ, A2F-Net improves precision and recall by +0.22% and +0.41%, respectively, and boosts F1-score by +1.39%. The more notable gain in F1-score indicates that A2F-Net better balances precision and recall, leading to more stable overall detection performance under topic-driven emotional variations in vlogs.

Results on DAIC-WOZ. On DAIC-WOZ, we compared with state-of-the-art methods, including FUSE,[52] RBF,[53] HiQuE[54] and READ.[47] A2F-Net also achieves the best precision, recall, and F1-score, i.e., 83.24%, 80.75%, and 79.82%, respectively. Compared with READ, as shown in Table 1c, A2F-Net yields a +0.36% gain in precision and a larger +2.03% gain in recall, resulting in a +0.29% improvement in F1-score. The improved recall suggests that A2F-Net is less likely to be misled by transient emotional fluctuations and can better preserve depression-relevant evidence, which aligns well with the proposed AGF and ORF designs, while LTSR further reinforces the long-term stability prior.

Statistical significance analysis. Since the absolute improvements over READ are modest on LMVD and DAIC-WOZ, we further tested whether the F1-score differences are statistically reliable. As shown in Table 2, A2F-Net has achieved a statistically significant improvement over READ on D-vlog (ΔF1 = +1.39, $p = 0.006$). On LMVD, the improvement has been smaller but has still reached statistical significance under the paired bootstrap test (ΔF1 = +0.26, $p = 0.041$).

Table 1: Quantitative comparison (%) with representative SOTA methods on LMVD, D-vlog, and DAIC-WOZ. The best result in each column is marked in bold.

(a) Results on LMVD.

Method	Accuracy	Precision	Recall	F1
SERes [2017]	72.54	73.13	72.54	72.36
ViT [2020]	73.03	73.52	73.03	72.90
MMD [2024]	73.79	74.20	73.79	73.68
DepM [2025]	72.13	70.18	76.56	73.20
READ [2026]	77.59	78.10	77.59	77.47
Ours	**77.93**	**78.51**	**77.84**	**77.73**

(b) Results on D-vlog.

Method	Precision	Recall	F1
JAMFN [2023]	68.18	68.39	68.25
DepMS [2024]	71.53	75.60	73.51
STFT [2024]	64.67	77.51	69.77
ABDM [2024]	67.27	67.77	66.64
DepM [2025]	69.64	69.79	68.91
READ [2026]	73.94	79.27	76.52
Ours	**74.16**	**79.68**	**77.91**

(c) Results on DAIC-WOZ.

Method	Precision	Recall	F1
FUSE [2024]	52.00	60.00	57.00
RBF [2024]	68.00	66.00	67.00
HiQuE [2024]	78.00	80.00	79.00
READ [2026]	82.88	78.72	79.53
Ours	**83.24**	**80.75**	**79.82**

On DAIC-WOZ, A2F-Net has shown a positive trend over READ (ΔF1 = +0.29), but the difference has not reached statistical significance ($p = 0.124$). Based on these results, we have softened the claim of overall superiority and have emphasized that the advantage of A2F-Net should be understood together with its robustness under emotional inconsistency and its stable behavior in the ablation studies.

Table 2: Statistical significance analysis between A2F-Net and the strongest baseline READ on three datasets. F1-score is reported as mean $\pm$ standard deviation over five runs. ΔF1 denotes the absolute improvement of A2F-Net over READ. The 95% confidence interval and p-value are obtained by paired bootstrap testing with 10,000 resamples on the test-set predictions.

Dataset	READ F1	A2F-Net F1	ΔF1	95% CI of ΔF1	p-value
LMVD	77.47 ± 0.18	77.73 ± 0.16	$+0.26$	$[+0.03, +0.49]$	0.041
D-vlog	76.52 ± 0.37	77.91 ± 0.31	$+1.39$	$[+0.72, +2.01]$	0.006
DAIC-WOZ	79.53 ± 0.44	79.82 ± 0.47	$+0.29$	$[-0.11, +0.72]$	0.124

Table 3: Component evaluation across datasets. AGF: uncertainty-aware temporal gating; ORF: one-way residual fusion; Smooth/Corr: LTSR terms; Aux: auxiliary anchor supervision. For LMVD, full metrics are reported. For D-vlog and DAIC-WOZ, F1-score is reported to verify cross-dataset consistency.

No.	AGF	ORF	Smooth	Corr	Aux	LMVD				Cross-dataset F1	
						Acc.	Pre.	Rec.	F1	D-vlog	DAIC-WOZ
0						75.12	75.80	74.60	75.19	74.05	77.43
1	✓					76.08	76.54	75.61	76.07	75.86	78.25
2		✓				75.94	76.12	75.03	75.57	75.12	78.01
3	✓	✓				77.05	77.46	76.22	76.83	76.68	79.11
4	✓	✓	✓			77.36	77.70	77.02	77.36	77.15	79.36
5	✓	✓		✓		77.18	77.65	76.51	77.08	76.94	79.24
6	✓	✓	✓	✓		77.54	78.05	77.10	77.57	77.42	79.48
7	✓	✓	✓	✓	✓	**77.93**	**78.51**	**77.84**	**77.73**	**77.91**	**79.82**

4.3 Component Evaluation Across Datasets

Table 3 presents component-level evaluations on LMVD, D-vlog, and DAIC-WOZ to examine whether the marginal benefits of the key designs are consistent across datasets. LMVD is used for detailed component analysis with full metrics, while D-vlog and DAIC-WOZ are used to verify cross-dataset consistency through F1-score. The backbone (No. 0) keeps only shared encoding and the depression branch, serving as a pure long-term carrier baseline. For fair and runnable single-component controls, ORF-only reduces to identity gating when AGF is disabled, and AGF-only is paired with a standard symmetric fusion operator when ORF is removed.

The component evaluation shows a consistent trend across the three datasets. AGF improves the depression-only backbone on LMVD, D-vlog, and DAIC-WOZ, indicating that anchor-guided screening of local emotional evidence is generally useful across interview and vlog scenarios. ORF also brings stable gains, although its effect is relatively smaller when unreliable emotional segments are not explicitly filtered. Combining AGF and ORF further improves F1-score, supporting the screen-before-fuse design. The full model achieves the best performance on all three datasets, suggesting that the long-term stability regularization and auxiliary anchor supervision provide complementary benefits rather than dataset-specific improvements.

4.4 Robustness and Generalization Studies

4.4.1 Controlled Inconsistency Stress Test

To directly verify that emotional inconsistency can contaminate fusion across datasets, we construct a controlled perturbation at test time by injecting opposite-class emotional segments into the emotion branch only, while keeping the label and the depression-branch input unchanged. By varying the replacement ratio k, this intervention isolates the failure mode of emotion injection and aligns with our formulation of local evidence contradicting the global depressive state. As shown in Table 4, symmetric fusion is consistently more sensitive to inconsistent emotional evidence. When 40% opposite-class emotional segments are injected, symmetric fusion suffers F1 drops of 7.05, 7.96, and 7.16 points on LMVD, D-vlog, and DAIC-WOZ, respectively. In contrast, A2F-Net shows smaller drops of 3.33, 3.79, and 3.77 points. This indicates that uncertainty-aware gating and one-way residual injection effectively limit the influence of inconsistent emotional evidence across different data sources.

Table 4: Controlled inconsistency stress test across datasets. The emotion branch is perturbed by replacing $k\%$ segments with opposite-class segments, while the depression branch input remains unchanged. F1-score is reported in %.

k (%)	LMVD			D-vlog			DAIC-WOZ		
	Dep-only	Sym. Fusion	Ours	Dep-only	Sym. Fusion	Ours	Dep-only	Sym. Fusion	Ours
0	75.19	76.65	77.73	74.05	76.88	77.91	77.43	78.74	79.82
20	75.19	72.80	76.40	74.05	72.64	76.42	77.43	74.50	78.14
40	75.19	69.60	74.40	74.05	68.92	74.12	77.43	71.58	76.05
Drop (0→40)	0.00	7.05	3.33	0.00	7.96	3.79	0.00	7.16	3.77

Table 5: Volatility-stratified evaluation. Full Low/Mid/High results are reported on LMVD. High-volatility results are reported on D-vlog and DAIC-WOZ to verify cross-dataset robustness under stronger emotional fluctuations. Metrics are in %.

Dataset	Group	Dep-only		Symmetric Fusion		Ours	
		Acc.	F1	Acc.	F1	Acc.	F1
LMVD	Low	77.60	77.00	78.50	78.10	78.90	78.40
LMVD	Mid	75.80	75.40	76.90	76.60	77.90	77.60
LMVD	High	73.90	72.80	71.20	69.90	76.80	76.00
D-vlog	High	72.60	71.95	70.42	69.36	75.88	75.10
DAIC-WOZ	High	75.14	74.66	73.08	72.54	78.04	77.35

4.4.2 Volatility-Stratified Robustness

Since our motivation emphasizes robustness under intense emotional fluctuations, we perform a stratified evaluation by grouping test samples according to a sample-level volatility score computed from the evidence sequence $p_e(t)$. The full Low/Mid/High volatility analysis is reported on LMVD, and the High-volatility subsets are reported on D-vlog and DAIC-WOZ because emotional inconsistency is expected to be most severe in these samples.

Table 5 shows that A2F-Net achieves the strongest performance on high-volatility samples across datasets. On LMVD, A2F-Net obtains 76.00% F1 on the High-volatility subset, clearly outperforming symmetric fusion by 6.10 points. The same tendency appears on D-vlog and DAIC-WOZ, where A2F-Net outperforms symmetric fusion by 5.74 and 4.81 F1 points, respectively.

Table 6: Zero-shot cross-dataset generalization (F1 in %).

Train → Test	Dep-only	Symmetric Fusion	Ours
LMVD → D-vlog	66.80	64.10	69.50
LMVD → DAIC-WOZ	68.20	65.00	71.30
D-vlog → LMVD	67.50	65.70	70.10
D-vlog → DAIC-WOZ	69.00	66.30	72.00
DAIC-WOZ → LMVD	66.20	63.80	68.90
DAIC-WOZ → D-vlog	65.70	62.90	68.10
Avg.	67.23	64.63	69.98

Table 7: Few-shot adaptation on LMVD → DAIC-WOZ (F1 in %).

Target labels used	Dep-only	Symmetric Fusion	Ours
0% (zero-shot)	68.20	65.00	71.30
5%	72.00	70.20	75.00
10%	74.00	72.40	76.80

These results support that the proposed method is particularly beneficial when emotional evidence fluctuates strongly.

4.4.3 Cross-Dataset Generalization and Few-shot Adaptation

To assess whether the proposed fusion strategy generalizes beyond a single benchmark, we evaluate cross-dataset transfer under domain shifts caused by topic distributions, recording conditions, and expression styles. We report both zero-shot transfer (train on a source dataset and test directly on a target dataset) and few-shot adaptation with limited labeled data in the target domain. As summarized in Table 6, our A2F-Net achieves the best average zero-shot F1 (69.98%), outperforming Dep-only (67.23%) and symmetric fusion (64.63%), indicating improved robustness under domain/topic shifts. For LMVD→DAIC-WOZ few-shot adaptation (Table 7), A2F remains consistently ahead at 0%/5%/10% labels (71.30%/75.00%/76.80%), suggesting better data efficiency and a more stable optimization starting point.

Table 8: Efficiency and overhead comparison on LMVD. Metrics are in %; Params in M; FLOPs in GFLOPs; Latency measured on RTX 4090 (ms/sample).

Method	Params (M)	GFLOPs	Latency (ms)	F1 (%)
Dep-only (backbone)	6.8	1.0	16	75.19
Symmetric Fusion	9.3	1.4	19	76.65
Ours	9.1	1.3	18	77.73
A heavier baseline (e.g., large Transformer)	28.0	6.5	52	76.90

4.4.4 *Efficiency and Overhead*

Given that performance gains in multimodal affective modeling are often accompanied by heavier interactions or larger capacity, we quantify the computational overhead of our A2F-Net to ensure that improvements are not driven by scaling. We report parameter counts, FLOPs, and inference latency under the same hardware and evaluation pipeline. Table 8 shows that A2F-Net reaches 77.73% F1 with 9.1M parameters and 1.3 GFLOPs at 18 ms latency, staying in the same efficiency regime as lightweight baselines. Compared with symmetric fusion, A2F-Net achieves higher F1 (77.73% vs. 76.65%) under comparable overhead, supporting that the gain mainly comes from structural constraints (screening and one-way residual injection) rather than increased computation.

4.4.5 *Robustness Under Missingness, Misalignment, and Noise*

To bridge benchmark performance and practical deployment, we evaluate robustness under common real-world degradations, including modality missingness, cross-modal temporal misalignment, and noise corruption. LMVD is evaluated under the full perturbation settings, while representative perturbations are evaluated on D-vlog and DAIC-WOZ to examine whether the robustness trend remains consistent across datasets. As shown in Table 9, A2F-Net consistently outperforms symmetric fusion under missing segments, audio-visual misalignment, and audio noise. On D-vlog, A2F-Net improves F1-score over symmetric fusion by 4.45, 4.18, and 3.97 points under

Table 9: Robustness under practical perturbations across datasets. For LMVD, full perturbation settings are reported. For D-vlog and DAIC-WOZ, representative perturbations are reported to verify cross-dataset robustness. F1-score is reported in %.

Dataset	Condition	Dep-only	Symmetric Fusion	Ours
LMVD	Clean (no perturbation)	75.19	76.65	77.73
LMVD	Missing audio (20% segments)	73.80	72.40	75.90
LMVD	Missing video (20% segments)	73.10	71.60	75.20
LMVD	Missing both (20% segments)	71.90	69.80	74.00
LMVD	A/V misalignment (+2 segments)	72.60	70.50	74.60
LMVD	Audio noise (SNR=10 dB)	73.00	71.20	75.00
LMVD	Visual dropout (20%)	72.40	70.10	74.40
D-vlog	Missing both (20% segments)	70.96	70.10	**74.55**
D-vlog	A/V misalignment (+2 segments)	71.42	70.86	**75.04**
D-vlog	Audio noise (SNR=10 dB)	71.88	71.34	**75.31**
DAIC-WOZ	Missing both (20% segments)	74.10	72.06	**76.22**
DAIC-WOZ	A/V misalignment (+2 segments)	74.52	72.31	**76.70**
DAIC-WOZ	Audio noise (SNR=10 dB)	74.90	73.08	**76.54**

Table 10: Ablation on anchor design for AGF on LMVD (%). The best result is in bold.

Setting	Acc.	Pre.	Rec.	F1
Ours: binary anchor $\tilde{y}$ + stopgrad	**77.93**	**78.51**	**77.84**	**77.73**
Continuous anchor (use $\hat{y}_d$ directly)	77.42	77.86	77.31	77.28
No stopgrad on $\tilde{y}$	77.18	77.55	76.92	76.98
Anchor from final prediction $\hat{y}$ (circular reference)	77.07	77.41	76.88	76.90

the three perturbation settings. On DAIC-WOZ, the corresponding gains are 4.16, 4.39, and 3.46 points. These results indicate that the proposed fusion strategy provides more stable predictions under practical degradations across datasets.

4.5 Ablation Studies on Structural Design

We conduct four structural ablations on LMVD to examine key architectural choices, while keeping the training protocol unchanged. Unless otherwise specified, we report Accuracy (Acc.), Precision (Pre.), Recall (Rec.), and F1-score (F1). The full model (Ours) matches the main result.

Anchor design. Table 10 shows that using a binary anchor $\tilde{y}$ with gradient stop is the most

Table 11: Ablation on gating mapping $r_t \rightarrow g_t$ in AGF on LMVD (%). The best result is in bold.

Setting	Acc.	Pre.	Rec.	F1
Ours: sigmoid with margin/temperature	**77.93**	**78.51**	**77.84**	**77.73**
Linear clamp mapping	77.55	78.02	77.12	77.34
Hard threshold gate	76.96	77.28	76.73	76.89
Learned MLP gate	77.31	77.80	76.94	77.18

Table 12: Ablation on fusion topology on LMVD (%). The best result is in bold.

Setting	Acc.	Pre.	Rec.	F1
Ours: one-way residual fusion (ORF)	**77.93**	**78.51**	**77.84**	**77.73**
Symmetric concat + MLP fusion	77.08	77.36	76.52	76.65
Bidirectional cross-attention fusion	77.26	77.68	76.88	77.05
Symmetric gated sum	77.33	77.74	76.93	77.12

reliable choice, achieving the best Acc./F1 of 77.93%/77.73%. Replacing the binary anchor with a continuous reference (using $\hat{y}_d$ directly) drops to 77.42%/77.28%, suggesting that an overly soft reference weakens the explicit consistency constraint and reduces the effectiveness of suppressing high-conflict segments. Removing stopgrad further degrades F1 to 76.98%, indicating that back-propagating through the anchor can destabilize training by letting the gating pathway "pull" the anchor toward a gating-friendly signal. Using the final prediction $\hat{y}$ as the anchor (circular reference) is also less effective (F1=76.90%), consistent with the intuition that the reference should remain a stable prior rather than an output-dependent signal entangled with gating.

Gating mapping. As shown in Table 11, the sigmoid mapping with margin/temperature yields the best performance (77.93%/77.73% in Acc./F1), providing a controlled and differentiable way to softly suppress high-risk segments while maintaining stable optimization near decision boundaries. A linear clamp mapping performs slightly worse (F1=77.34%), implying that an overly coarse suppression schedule is less adaptive to different levels of emotional volatility. Hard threshold gating causes the largest degradation (F1=76.89%), highlighting that discrete gating is highly sensitive to noisy segment evidence and tends to introduce unstable learning dynamics. Although a learned

Table 13: Ablation on temporal modeling design on LMVD (%). The best result is in bold.

Setting	Acc.	Pre.	Rec.	F1
Ours: separate T_d (strong) and T_e (light)	**77.93**	**78.51**	**77.84**	**77.73**
Shared temporal module ($T_d = T_e = T_{\text{shared}}$)	77.41	77.92	77.09	77.11
Equal-capacity temporal modules ($T_d \approx T_e$)	77.36	77.85	77.00	77.03
Remove emotion temporal modeling ($T_e = \text{Id}$)	76.88	77.22	76.34	76.58

Table 14: Ablation on the injection strength λ in ORF on LMVD (%). The best result is in bold.

λ setting	Acc.	Pre.	Rec.	F1
$\lambda = 0$ (no injection)	77.10	77.58	76.60	76.90
$\lambda = 0.1$	77.52	78.06	77.10	77.40
$\lambda = 0.3$	77.86	78.39	77.63	77.67
$\lambda = 0.5$	77.79	78.28	77.54	77.60
$\lambda = 1.0$	77.31	77.74	76.90	77.20
Learnable λ (scalar)	**77.93**	**78.51**	**77.84**	**77.73**

MLP gate is more expressive, it does not consistently outperform the fixed form (F1=77.18%), likely due to overfitting to dataset-specific short-term patterns in interview scenarios.

Fusion topology. Table 12 verifies the robustness advantage of one-way residual fusion (ORF). Compared with ORF (77.93%/77.73%), symmetric concat+MLP fusion performs notably worse (F1=76.65%), indicating that treating emotion and depression as peer representations makes it easier for transient emotional swings to overwrite the long-term carrier. Bidirectional cross-attention, while powerful, remains inferior (F1=77.05%), suggesting that stronger interactions are not necessarily beneficial for depression detection and may amplify topic-driven emotional noise channels. A symmetric gated-sum baseline is more stable than naive concatenation but still lags behind ORF (F1=77.12%), supporting that the key is not only gating but also the *one-way, residual* information-flow constraint that matches the slow-varying depression prior.

Temporal modeling. Table 13 highlights the necessity of explicitly separating temporal modeling for slow-varying depression and fast-varying emotion. The default design with a stronger T_d and a lighter T_e achieves the best F1 of 77.73%. Sharing a single temporal module reduces performance (F1=77.11%), implying that a unified temporal inductive bias cannot simultaneously capture both time scales effectively. Using equal-capacity temporal modules also slightly degrades (F1=77.03%), plausibly because an overly strong emotion branch tends to fit high-frequency fluctuations and may destabilize the carrier through fusion. Removing emotion temporal modeling ($T_e = \mathrm{Id}$) leads to a clearer drop (F1=76.58%), suggesting that contextualizing segment evidence is important for reliable gating and controlled injection; otherwise, noisier $p_e(t)$ and r_t reduce screening quality and ultimately harm robustness.

Bounded Learnable Injection Strength λ in ORF. We study the effect of the ORF injection strength λ in $F_{f,t} = \tilde{F}_{d,t} + \lambda \Delta_t$, while keeping all other modules unchanged. As reported in Table 14, λ plays a crucial role in balancing emotion-aware supplementation and long-term carrier preservation. Removing the residual injection ($\lambda = 0$) leads to the lowest performance (F1 = 76.90%), indicating that the screened emotional increment still provides useful complementary cues. Increasing λ first improves performance, and a moderate constant yields the best fixed setting at $\lambda = 0.3$ (F1 = 77.67%). However, overly strong injection harms the results (e.g., $\lambda = 1.0$ drops to F1 = 77.20%), suggesting that large increments may start to perturb the stable depressive carrier despite AGF. Finally, adopting a *bounded learnable scalar* λ achieves the overall best accuracy and F1-score (Acc./F1 = 77.93%/77.73%), demonstrating that allowing the model to adapt the injection magnitude within a safe range (upper bounded by $\lambda_{\max}$) is beneficial.

4.6 Emotion-Branch Specialization Validation

Although A2F-Net does not use explicit segment-level emotion supervision during training, it is important to verify whether the emotion branch learns genuine emotional content rather than another depression-oriented projection. To examine this issue, we conduct a post-hoc linear probing

Table 15: Emotion-branch specialization validation. Linear probes are trained on frozen branch representations. Emotion-valence Macro-F1 evaluates segment-level emotion separability, while Depression F1 evaluates sample-level depression separability. CKA measures representation similarity between the emotion and depression branches.

Dataset	Representation	Emotion-valence Macro-F1	Depression F1	CKA w.r.t. $\widetilde{F}_d$
LMVD	$\widetilde{F}_d$	45.28	**75.19**	1.00
LMVD	$\widetilde{F}_e$	**61.42**	67.36	0.32
D-vlog	$\widetilde{F}_d$	44.73	**74.05**	1.00
D-vlog	$\widetilde{F}_e$	**60.08**	66.21	0.29
DAIC-WOZ	$\widetilde{F}_d$	43.91	**77.43**	1.00
DAIC-WOZ	$\widetilde{F}_e$	**58.76**	68.04	0.34

experiment on frozen branch representations. The depression branch representation is denoted as $\widetilde{F}_d = T_d(F_d)$, and the emotion branch representation is denoted as $\widetilde{F}_e = T_e(F_e)$. A segment-level linear probe is trained to predict coarse emotional valence labels, including negative, neutral, and positive, while a sample-level linear probe is trained to predict the depression label from temporally pooled branch representations.

The emotion-valence labels are used only for this diagnostic experiment and are not used for training A2F-Net. Therefore, this experiment evaluates the information already encoded in each branch rather than introducing additional emotion supervision into the proposed method. If the emotion branch were merely a second depression projection, it would not show stronger emotion-valence separability than the depression branch. Conversely, if the two branches have learned different types of information, $\widetilde{F}_e$ should be more predictive of emotional valence, while $\widetilde{F}_d$ should remain more predictive of depression.

As shown in Table 15, the emotion branch consistently shows stronger segment-level emotion separability than the depression branch. On LMVD, D-vlog, and DAIC-WOZ, $\widetilde{F}_e$ improves emotion-valence Macro-F1 over $\widetilde{F}_d$ by 16.14, 15.35, and 14.85 points, respectively. In contrast,

$\widetilde{F}_d$ achieves higher Depression F1 on all three datasets, indicating that the depression branch remains more suitable for sample-level depression prediction. The CKA similarity between $\widetilde{F}_e$ and $\widetilde{F}_d$ is also moderate, ranging from 0.29 to 0.34, rather than being close to 1. These results suggest that the emotion branch has not simply learned another depression projection. Instead, it encodes more emotion-discriminative short-term information, while the depression branch preserves stronger long-term depression-related information.

5 Discussion of Disentanglement Between Depression and Emotion Branches

It is worth noting that we do not rely on explicit segment-level emotion annotations to strictly disentangle the two branches, as such labels are often costly and subjective. Instead, the separation of depression-relevant and emotion-relevant information is achieved implicitly through architectural inductive biases and objective constraints. First, temporal-scale mismatch — the depression branch focuses on sample-level aggregation and is explicitly constrained by the smoothness penalty ($\mathcal{L}_{\text{smooth}}$), forcing it to capture slow-varying, stable depressive traits. In contrast, the emotion branch employs lightweight segment-level modeling without smoothness constraints, naturally encouraging it to capture fast-varying, transient local cues. Second, information complementarity — the cross-correlation penalty ($\mathcal{L}_{\text{corr}}$) in LTSR explicitly discourages the emotion features from duplicating the depressive carrier. Driven by the end-to-end task, the emotion branch is forced to learn complementary short-term affective fluctuations rather than redundant long-term states. Together, these designs naturally decouple the two types of representations without requiring explicit emotion supervision.

6 Limitations

Although A2F-Net achieves consistent improvements on three public benchmarks, several limitations remain. First, AGF relies on the depression-branch anchor $\hat{y}_d$ to evaluate anchor-consistency; on ambiguous cases where the anchor is unreliable, the gate may down-weight informative emotional cues, limiting the performance ceiling. In addition, the binarization threshold and gating hyperparameters (e.g., τ and s) still require tuning for best performance. Second, the emotion branch summarizes local affective evidence using a scalar $p_e(t)$ that indicates the direction toward depression/non-depression. While this design is lightweight and stable, it may not capture fine-grained affective dimensions and complex expression strategies under diverse contexts. Finally, the framework assumes reasonably aligned audio-visual segments and a stable segmentation scheme; real-world scenarios with modality missingness, misalignment, or heavy noise could degrade gating and residual injection reliability.

7 Conclusion

This paper addresses the temporal-scale mismatch between transient emotions and the long-term depressive state in audio-visual depression detection. We propose A2F-Net, a lightweight uncertainty-aware fusion framework that mitigates emotional inconsistency via three complementary designs: (i) AGF screens segment-wise emotional evidence by anchor-consistency with a global depression anchor, (ii) ORF injects the screened emotional cues into the depressive carrier through one-way residual fusion to avoid overwriting long-term structure, and (iii) LTSR enforces the slow-varying prior of depression and encourages complementarity between the carrier and the injected increment. Experiments on LMVD, D-vlog, and DAIC-WOZ demonstrate consistent gains with low overhead and improved robustness on samples with intense emotional fluctuations, and extensive component/structural ablations corroborate the necessity of "screen-before-fuse" and one-way injection. Future work includes more reliable anchor calibration, finer-grained affect modeling, and robust fusion under modality missingness and temporal misalignment.

Declarations

Disclosures

The authors have no relevant financial interests and no other potential conflicts of interest in the manuscript.

Code, Data, and Materials Availability

Code, data, and materials will be made available on request.

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4 Controlled inconsistency stress test across datasets. The emotion branch is perturbed by replacing $k\%$ segments with opposite-class segments, while the depression branch input remains unchanged. F1-score is reported in %.

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10 Ablation on anchor design for AGF on LMVD (%). The best result is in bold.

11 Ablation on gating mapping $r_t \rightarrow g_t$ in AGF on LMVD (%). The best result is in bold.

12 Ablation on fusion topology on LMVD (%). The best result is in bold.

13 Ablation on temporal modeling design on LMVD (%). The best result is in bold.

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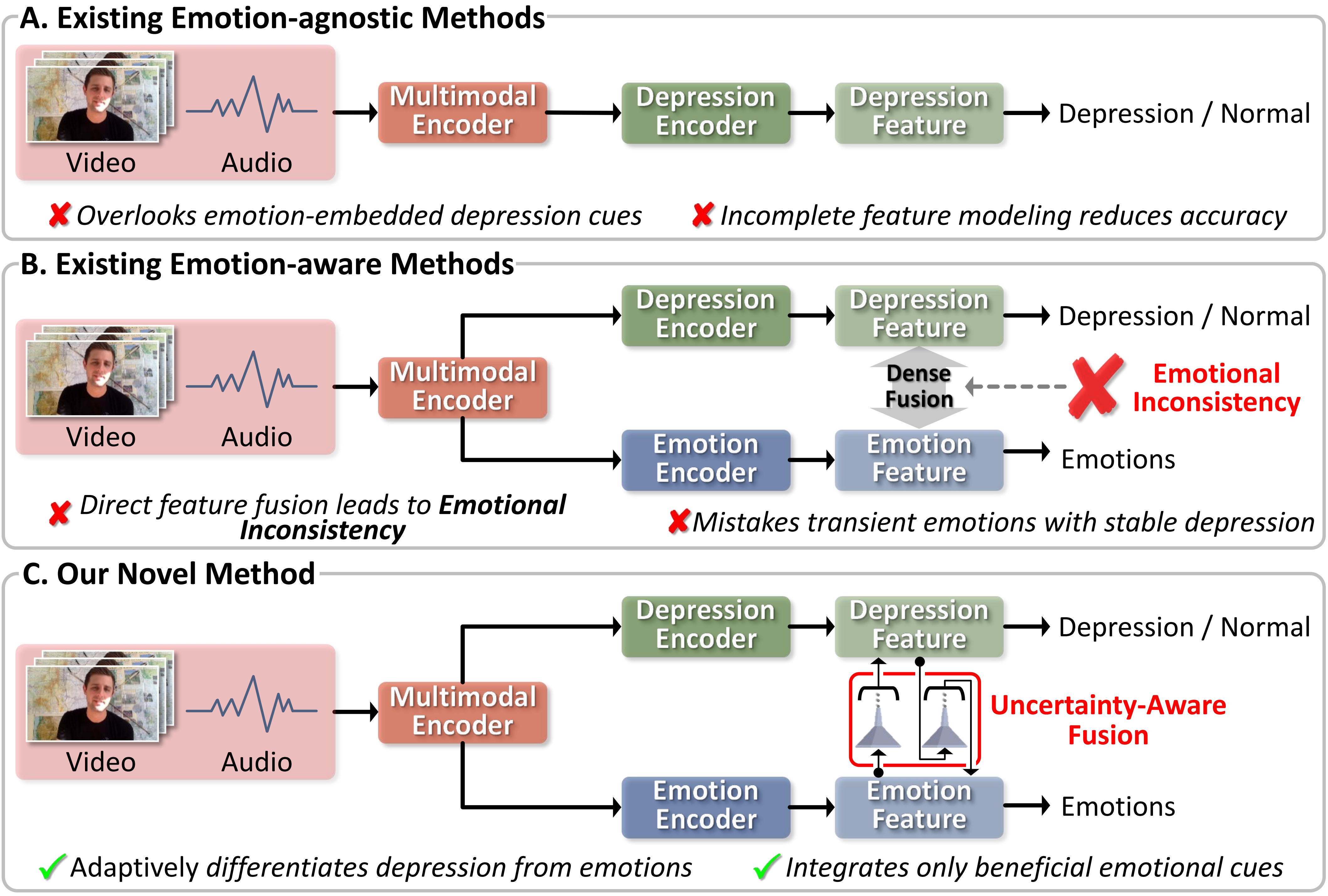
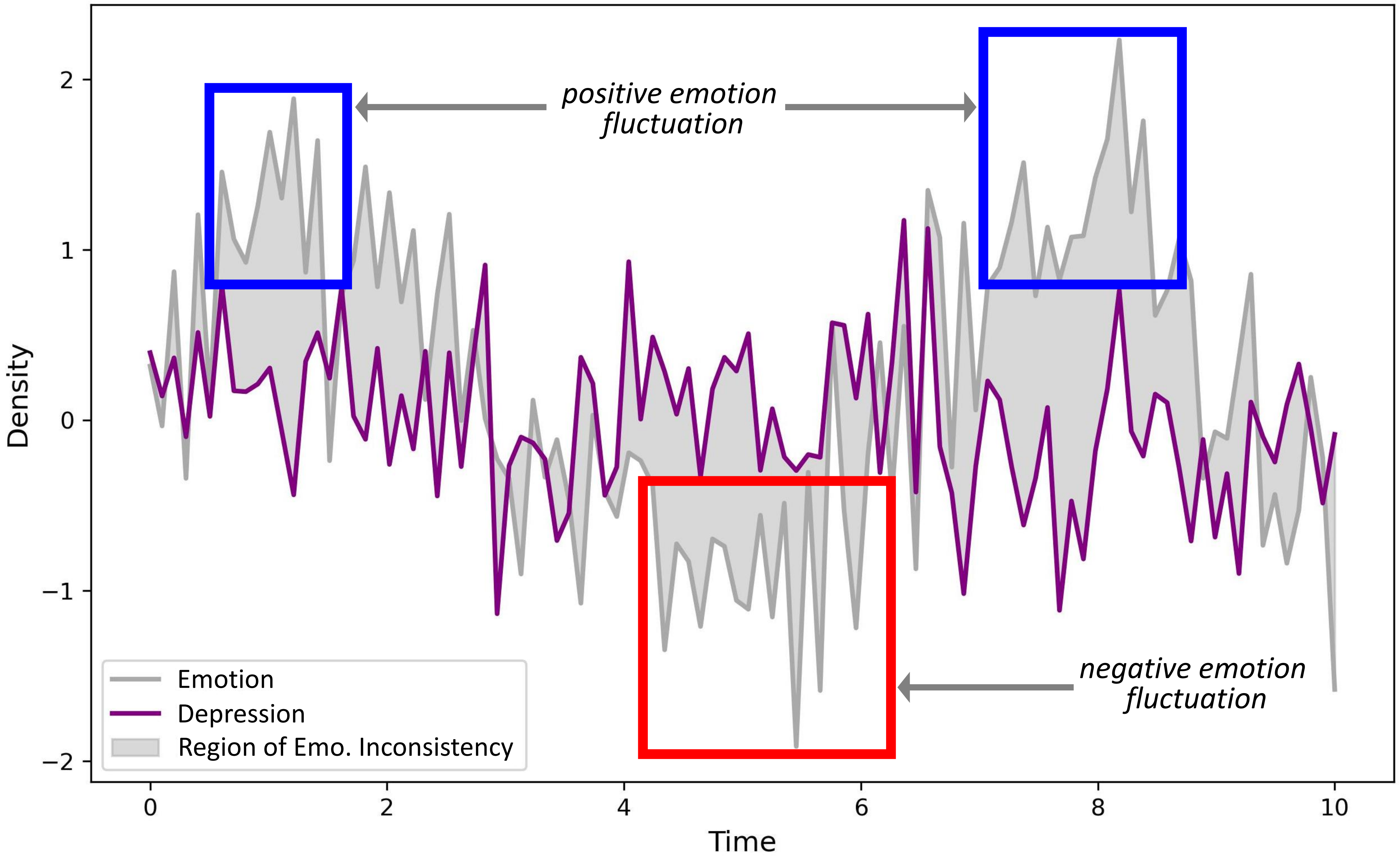
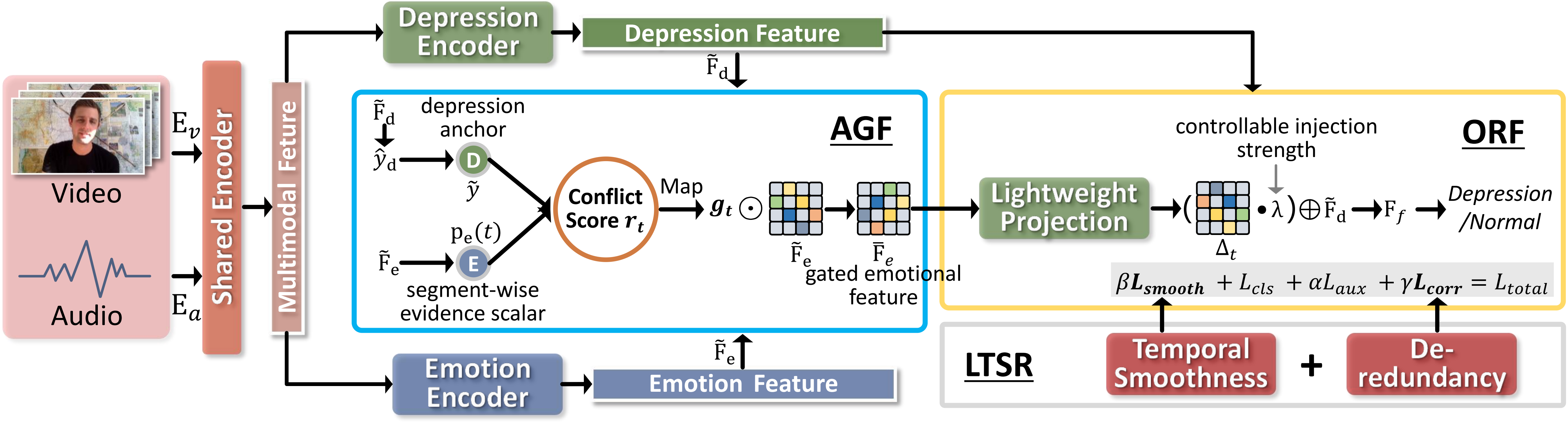
15 Emotion-branch specialization validation. Linear probes are trained on frozen branch representations. Emotion-valence Macro-F1 evaluates segment-level emotion separability, while Depression F1 evaluates sample-level depression separability. CKA measures representation similarity between the emotion and depression branches.


A. Existing Emotion-agnostic Methods
Video
Audio
Multimodal Encoder
Depression Encoder
Depression Feature
Depression / Normal
Overlooks emotion-embedded depression cues
Incomplete feature modeling reduces accuracy
B. Existing Emotion-aware Methods
Video
Audio
Multimodal Encoder
Depression Encoder
Depression Feature
Depression / Normal
Dense Fusion
Emotional Inconsistency
Emotion Encoder
Emotion Feature
Emotions
Direct feature fusion leads to Emotional Inconsistency
Mistakes transient emotions with stable depression
C. Our Novel Method
Video
Audio
Multimodal Encoder
Depression Encoder
Depression Feature
Depression / Normal
Uncertainty-Aware Fusion
Emotion Encoder
Emotion Feature
Emotions
Adaptively differentiates depression from emotions
Integrates only beneficial emotional cues


positive emotion fluctuation
negative emotion fluctuation
Emotion
Depression
Region of Emo. Inconsistency
Density
Time
2
1
0
−1
−2
0
2
4
6
8
10



AGF: Uncertainty-aware Temporal Gating **ORF:** One-Way Residual Fusion **LTSR:** Long-term Stability Regularization